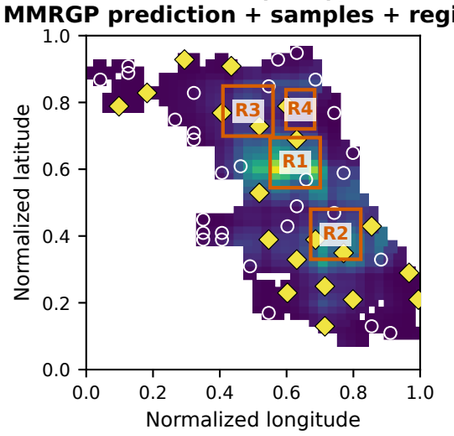
Crimes case study: final 50-cell designs, MMRGP surfaces, and four recommended regions

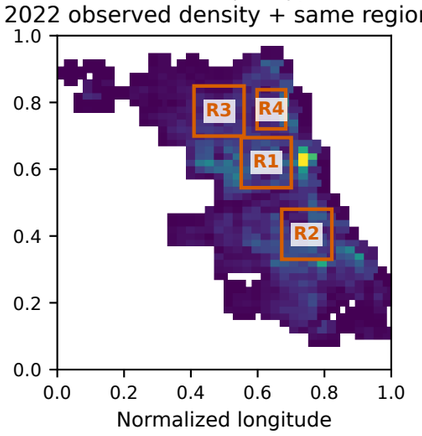
Initial 30 cells · Acquired 20 cells · Strict-valid GSO regions R1–R4

KARMA (final)†
MMRGP prediction + samples + regions



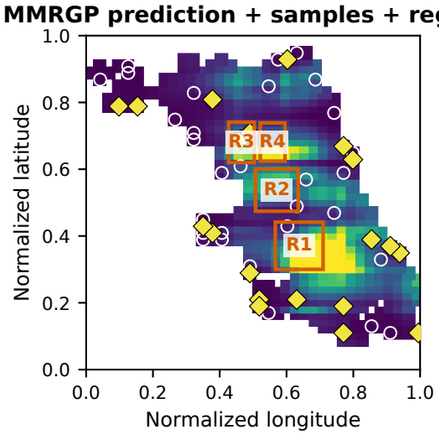
Dual-channel knowledge + ROI arbitration

KARMA (final)†
2022 observed density + same regions



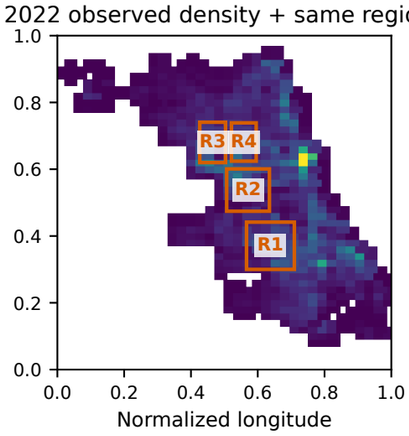
RP=0.50 CRP=0.50 Lift=1.031 TopN=0.387

C1: Maximin-LHS
MMRGP prediction + samples + regions



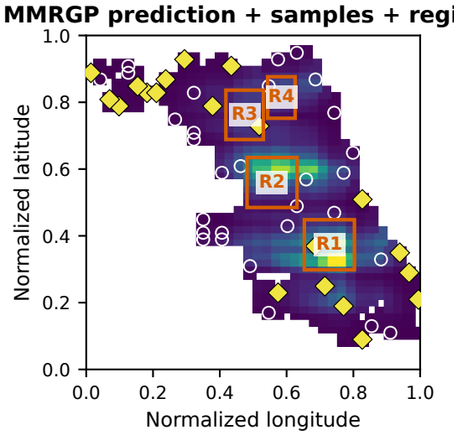
Space-filling, no feedback

C1: Maximin-LHS
2022 observed density + same regions



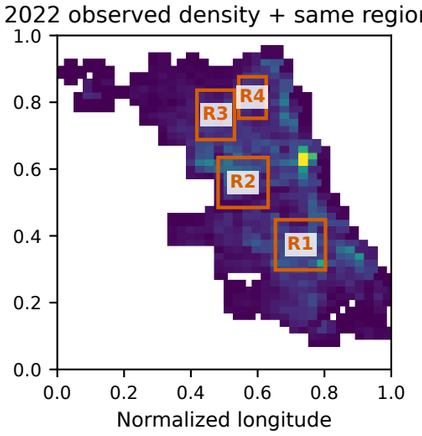
RP=0.50 CRP=0.50 Lift=1.154 TopN=0.255

C2: N-MeanMSE
MMRGP prediction + samples + regions



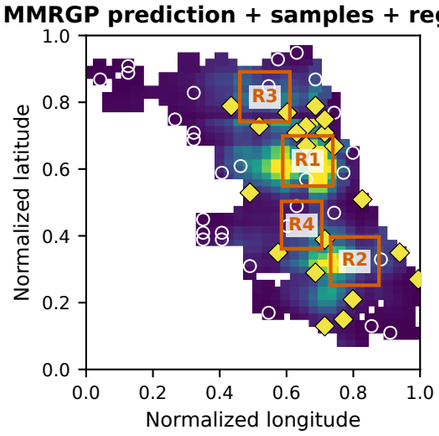
Global pointwise uncertainty

C2: N-MeanMSE
2022 observed density + same regions



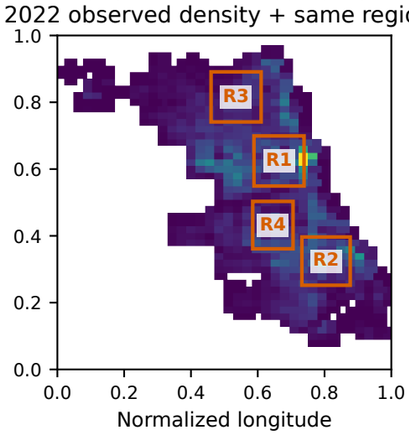
RP=0.25 CRP=0.25 Lift=0.997 TopN=0.312

C10: Voronoi-MixedMetric
MMRGP prediction + samples + regions



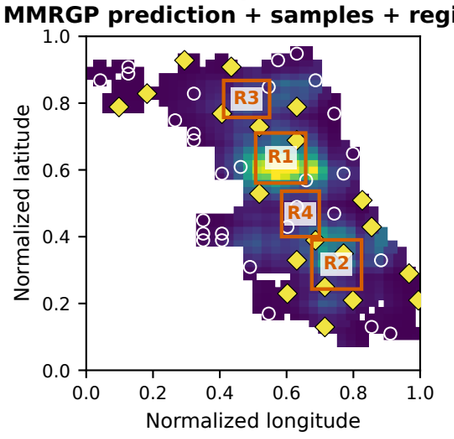
Data-driven regional sampling

C10: Voronoi-MixedMetric
2022 observed density + same regions



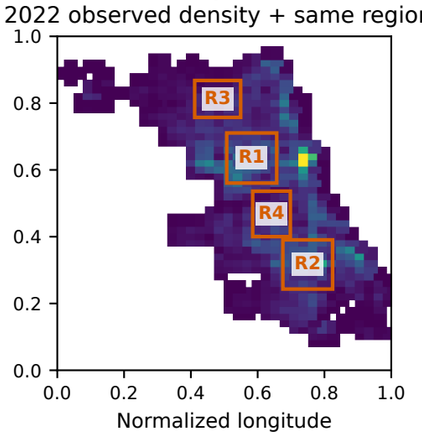
RP=0.50 CRP=0.50 Lift=1.044 TopN=0.348

ROI-D
MMRGP prediction + samples + regions



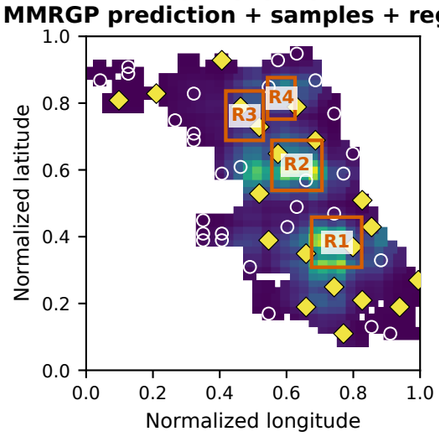
Same-framework no-knowledge ablation

ROI-D
2022 observed density + same regions



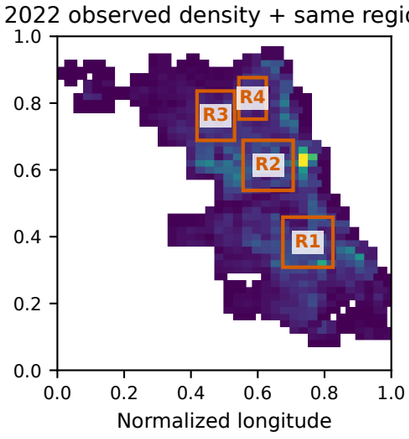
RP=0.50 CRP=0.50 Lift=0.992 TopN=0.324

C7: πBO-ROI
MMRGP prediction + samples + regions

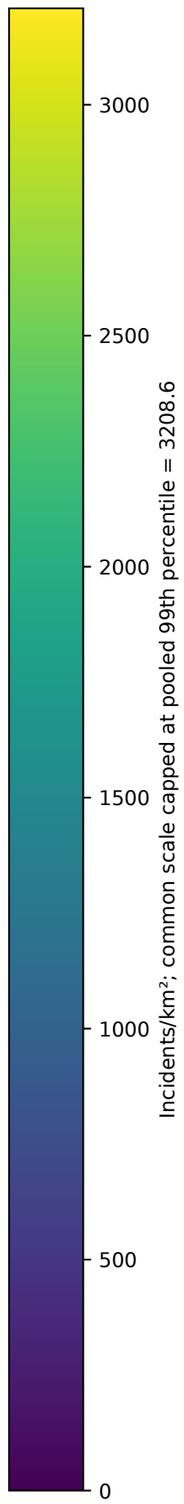


Pointwise fixed prior inside data ROI

C7: πBO-ROI
2022 observed density + same regions



RP=0.50 CRP=0.50 Lift=1.012 TopN=0.346



Only the 659 valid 2022 grid cells are shown; no spatial interpolation. Values above the color cap are retained in the data.
†KARMA uses a 2022 target-year prior; seed 172 is one predeclared diagnostic run, not the ten-seed average.